Copying, forgetting and evidence: a mean-field theory of the reading reversal and of persistence in swarms of Bayesian agents

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Abstract

The system. A swarm of $N = 24$ exact Bayesian reasoners shares a public board. Each agent reads a few recent posts, forgets what it read at a constant rate, votes, and reposts facts. A covert controller injects true but selected facts that favour a wrong answer, and then stops. The swarm shows two robust phenomena:

1. a *reading reversal*: reading more helps the controller when memory is short and protects the swarm when memory is long;
2. *persistence*: the controller's effect outlives its posting in one task and fades in another.

The theory. We coarse-grain the microscopic stochastic process into a small set of mean-field equations for fact densities in memories and on the board. The exact Bayesian inference is replaced by a logistic response to evidence counts. For every equation we state:

- the exact microscopic quantity it replaces;
- the closure that turns it into a closed equation, in what sense that closure is mean-field, and which way it biases the result.

What the theory explains.

- **The reversal** is a competition between the controller's net log-odds displacement and the population's falling sensitivity, as reading accumulates the agents' own evidence.
- **Persistence** is governed by a two-stage epidemic-type threshold $\mathcal{R}_C$ for controller facts.
- **Its magnitude** is set by finite-size extinction of facts. The relevant system size is copies per fact, and the swarm's widely spread evidence collapses while the controller's concentrated evidence survives.
- **Fading** can arise from frequency-dependent selection among circulating controller facts, driven by support-based reposting. This selection obeys an exact but unclosed Price equation.

We separate what was predicted in advance from what was found after seeing the data, and list what remains unexplained. The common framework was agreed with an independent agent (ChatGPT); its exact results are included with attribution.

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1 Introduction

The system is a minimal model of opinion manipulation among reasoning agents. The agents are not language models here: each is replaced by an exact Bayesian reasoner over a finite world of possible solutions to a logic task. This makes the microscopic dynamics fully specified and the posteriors computable. Two tasks are used:

- **task003,** an allocation puzzle;
- **Object Placements (OP),** a belief-tracking puzzle.

Each comes in two setups:

- **symmetric:** the agents' facts jointly prove the right answer;
- **no-proof:** no subset of their facts proves it.

The two phenomena are robust across tasks and setups.

(I) **Reading reversal.** Let $G(q)$ be the controller's effect on the share of wrong ($A2$) votes when agents read q posts per action, and $D(q_1 \to q_2) = G(q_2) - G(q_1)$. Then $D > 0$ at short memory and $D < 0$ at long memory, and the crossover memory falls as q grows.

(II) **Persistence.** After the controller stops at $T = 30$, its effect stays nearly flat for 100 time units in task003, but falls by 30–70% in OP. Freezing the board sustains the effect; cutting all communication removes it.

The aim is understanding, not numerical reproduction. We look for the smallest dynamics that exhibits both phenomena, and whose mechanism we can state. Its parameters are fixed before any run and are not fitted to the simulator. The models below never call the Bayesian engine. The engine is used only as a measuring instrument on stored simulator memories.

Plan.

- Section 2: the microscopic process.
- Section 3: the coarse-graining strategy and its scaling limit.
- Sections 4–7: each equation derived in turn.
- Section 8: what the closed system implies, analytically.
- Section 9: finite size.
- Section 10: the heterogeneity of the evidence response.
- Section 11: fading.
- Section 12: the independent agent's alternative closure.
- Section 13: the common framework and its status.

2 The microscopic process

Simulation 2 is a continuous-time Markov jump process (code: `task003_llm_free/simulation_2/`).

State.

- Each agent $i = 1, \ldots, N$ carries a binary occupation vector $n_{if} \in \{0, 1\}$ over the facts f of the task world: $n_{if} = 1$ if agent i currently remembers fact f (its *active* facts). At $t = 0$ each agent holds exactly one of its own (native) facts.
- The *board* is an append-only sequence of posts. Each post carries one fact and its author.
- Each agent has a last vote $v_i \in \{A0, A1, A2\}$.

Events.

1. **Forgetting.** Each held fact has an independent exponential lifetime with rate $\gamma = -\ln \rho$. So ρ is the probability that a fact survives one time unit without being read again. Reading a held fact again restarts its lifetime.
2. **Agent action**, at the events of an independent Poisson clock of rate 1 for each agent:
   (a) *Read.* Form the window of the $W = 48$ most recent posts, after excluding the agent's own posts. Draw q of them uniformly without replacement. Set $n_{if} = 1$ for every fact read, starting a fresh lifetime.

(b) *Vote.* Compute the exact posterior $P(A_k \mid \text{active facts})$ over the three answers. Vote A_k with probability $P(A_k \mid \cdot)$. This is *probability matching*.

(c) *Post.* Among held facts, find those that *support the vote in context*: removing the fact from the agent's active set would lower the posterior of the answer voted for. Post one of them, chosen uniformly, preferring facts not read in this action. If none qualifies, fall back to facts that raise that posterior when held alone. If none does, post nothing.

3. **Controller.** Until $T = 30$, at rate $\lambda_0 = 1$, the controller reads part of the board and posts the facts from its pool (true facts favouring $A2$) that maximise $P(A2)$ given what it read. After T it is silent.

Observables. Every quantity is a controlled-minus-silent difference between two coupled arms of the process, with and without the controller:

- $\Delta_v(t)$: the gap in the mean last $A2$ vote. The simulator's G and $\delta(t)$ are Δ_v at and after T.
- $\Delta_p(t)$: the gap in the mean posterior $P(A2 \mid \text{memory})$ at a snapshot.
- $g(t) = \Phi_c(t) - \Phi_s(t)$: the gap in the probability of the next vote, just after reading.

Because last votes are refreshed at rate 1, $\dot{\Delta}_v = g - \Delta_v$. The mean-field model of Sections 3–7 computes g; its finite version measures Δ_v; the diagnostics of Section 11 measure Δ_p.

The only non-generic element of the process is the Bayesian posterior. Everything else (Poisson clocks, exponential forgetting, a finite first-in-first-out window) is standard stochastic-process material.

3 Coarse-graining strategy

The microscopic state space is enormous, because memories are subsets of up to about 40 facts and the board has a full history. We reduce it in three steps.

Approximation 1 (two exchangeable classes)**.** Facts are grouped into two classes: native facts A (F_A of them) and controller facts C (F_C). Within a class facts are treated as interchangeable: they have the same board share, the same holding probability and the same evidential weight.

Motivation. This is the minimal structure that distinguishes the swarm's evidence from the controller's. Its cost is that fact identity and combinatorial structure (proofs, context-dependent meaning) are discarded. Section 10 shows that this cost is real for the microscopic response.

Approximation 2 (inference replaced by a response function)**.** The exact posterior is replaced by a logistic function of the numbers of facts held in each class (Section 6).

Approximation 3 (mean field)**.** Every agent is replaced by the average agent, and the board by its average composition. Concretely:

- averages of products are replaced by products of averages;
- averages of nonlinear functions are replaced by the functions of averages.

The scaling limit behind Approximation 3. Take $N \to \infty$ and $W \to \infty$ with the ratio N/W and the class sizes F_α fixed. Then every fact has $O(N)$ copies in memories and $O(W)$ copies on the board, and the system is a density-dependent Markov chain in the densities

$$m_\alpha = \frac{1}{N} \sum_i n_{if} \ (f \in \alpha), \qquad b_\alpha = \frac{\#\{\text{class-}\alpha \text{ posts in the window}\}}{W}. \tag{1}$$

By the law of large numbers for such chains (Kurtz's theorem), the densities converge to the solution of deterministic rate equations, with Gaussian fluctuations of relative size $O(\Omega^{-1/2})$. Here Ω is the number of copies per fact, about Nm_α in memories and Wb_α/F_α on the board.

Two points follow.

- **A finite number of facts F_α does not prevent mean field.** The large parameter is copies per fact, not facts.
- **What mean field loses is exactly the finite-Ω physics:** fluctuations, and absorption at zero copies (extinction of a fact). In the simulator, Ω is between about 1.5 and 12 (Table 3), so these effects cannot be neglected. Section 9 restores them.

Two features of the simulator are not of density-dependent form and survive only as approximations: a reader cannot read its own posts, and the vote is a nonlinear function of a whole memory.

4 Memory: Equation 1

Exact single-site equation. Fix agent i and fact f of class α. The occupation $n = n_{if}$ is a two-state process.

- $1 \to 0$ at rate γ. Because the lifetime is exponential (memoryless), restarting it on re-reading has no effect on this rate.
- $0 \to 1$ at rate $R_{if}(t)$, the probability that one action of agent i reads f, times the action rate 1.

Hence, exactly,

$$\frac{d}{dt}\mathbb{E}[n_{if}] = \mathbb{E}\big[(1 - n_{if})\, R_{if}\big] - \gamma\, \mathbb{E}[n_{if}]. \tag{2}$$

R_{if} is random: it depends on how many copies of f sit in agent i's window. If k of its W posts carry f, then for reading without replacement

$$R = 1 - \binom{W-k}{q} \Big/ \binom{W}{q}. \tag{3}$$

Closure. Three replacements turn Eq. (2) into a closed equation.

1. **Decorrelation.** $\mathbb{E}[(1-n)R] \approx (1 - \mathbb{E}n)\, \mathbb{E}R$. This neglects the correlation between what an agent holds and what it can read. For example, its own posts are excluded from its window, and those posts are preferentially facts it holds.
2. **Function of the mean.** $\mathbb{E}[R(k)] \approx R(\mathbb{E}k)$. R is concave in k, so by Jensen's inequality $\mathbb{E}[R(k)] \leq R(\mathbb{E}k)$. The mean field therefore *overestimates* acquisition when board copy numbers fluctuate, most strongly when copies are few.
3. **Sampling with replacement.** With $\mathbb{E}k = W b_\alpha / F_\alpha$, sampling q posts independently gives

$$r_\alpha = 1 - \Big(1 - \frac{b_\alpha}{F_\alpha}\Big)^q. \tag{4}$$

   For $q \ll W$ this differs little from Eq. (3). At $W = 48$, $q = 6$ and $k = 4$ the values are 0.407 (Eq. 4) and 0.425 (exact).

With $m_\alpha = \mathbb{E}[n_{if}]$, the result is

$$\boxed{\dot{m}_\alpha = (1 - m_\alpha)\, r_\alpha - \gamma\, m_\alpha} \tag{5}$$

Physical reading. Eq. (5) is a Langmuir adsorption–desorption equation: sites (memory slots) are filled from a reservoir (the board) at rate r and emptied at rate γ. On a fixed board its solution relaxes at rate $r + \gamma$ to the Langmuir isotherm

$$m^*_\alpha = \frac{r_\alpha}{r_\alpha + \gamma}. \tag{6}$$

It is linear in reading when memory is short ($m^* \approx r/\gamma$ for $\gamma \gg r$) and saturates at 1 when memory is long ($\gamma \to 0$).

5 Board: Equation 2

Exact bookkeeping. Let n_α be the number of class-α posts in the window, so that $b_\alpha = n_\alpha / W$. A new post enters the window and the oldest leaves:

$$n_\alpha \to n_\alpha + \mathbf{1}[\text{new} \in \alpha] - \mathbf{1}[\text{oldest} \in \alpha]. \tag{7}$$

Agents post at total rate N, assuming every action produces a post. Each post is class α with probability S_α (Section 7). The controller posts class C at rate $\lambda(t)$. Writing $u_C = 1$ and $u_A = 0$,

$$\frac{d\,\mathbb{E}n_\alpha}{dt} = N S_\alpha + \lambda u_\alpha - (N + \lambda)\, \Pr[\text{oldest} \in \alpha]. \tag{8}$$

Closure. The window is first-in-first-out (FIFO). The departing post is therefore the one that arrived W posts earlier, about $\tau = W/(N + \lambda)$ time units ago. Exactly, then, $\Pr[\text{oldest} \in \alpha]$ is the class composition of posts made at time $t - \tau$: a delay. The mean-field closure replaces it by the current composition, $\Pr[\text{oldest} \in \alpha] \approx b_\alpha$:

$$\boxed{\dot{b}_\alpha = \frac{N}{W}\,(S_\alpha - b_\alpha) + \frac{\lambda(t)}{W}\,(u_\alpha - b_\alpha)} \tag{9}$$

In what sense this is mean-field. Eq. (9) is *exact in expectation* for a different board, one on which each new post replaces a uniformly random old post (a Moran board). The approximation swaps FIFO removal for random removal. Without the controller, the solution is an exponentially weighted memory of past posting,

$$b_\alpha(t) = \int_0^\infty \frac{N}{W}\, e^{-Ns/W}\, S_\alpha(t-s)\, ds, \tag{10}$$

whereas the FIFO window is a boxcar average over the last τ (Fig. 1).

Both boards have the same turnover and identical stationary states, so the fixed points and linear thresholds of Section 8 do not depend on the choice. They differ in three ways:

- transient dynamics;
- the variance of post lifetimes, which is zero for FIFO and maximal for random removal. This matters for extinction (Section 9);
- behaviour when the posting rate changes. A count-based window slows down when agents abstain; the model assumes every action posts.

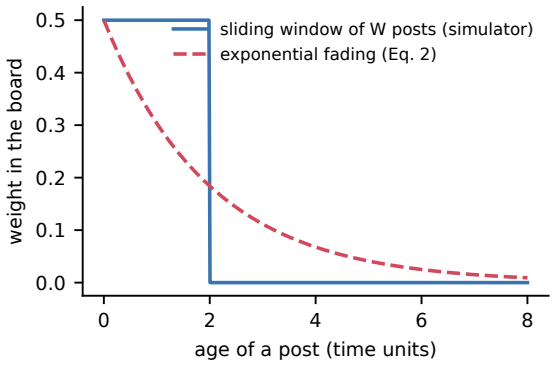


Figure 1: Board memory kernels with the same turnover. The simulator's FIFO window (solid) and the mean-field random-replacement board of Eq. (9) (dashed).

6 Vote: Equation 3

Exact log-odds update. Lump the two other answers into "not $A2$" and write $x(M) = \ln[P(A2 \mid M)/P(\neg A2 \mid M)]$. Adding one fact f to memory M gives, by Bayes' rule,

$$x(M \cup f) = x(M) + \theta_f(M), \qquad \theta_f(M) = \ln \frac{P(f \mid A2, M)}{P(f \mid \neg A2, M)}. \tag{11}$$

This is exact. $\theta_f(M)$ is the conditional log-likelihood ratio of f given what is already known.

Closure. Two approximations lead to the vote equation.

1. **Context-free evidence:** $\theta_f(M) = \theta_f$. This holds if the facts are conditionally independent given $A2$ and given $\neg A2$ ("naive Bayes"). Even conditional independence given each of the three answers is not enough, because the lumped likelihood $P(f \mid \neg A2, M)$ is a mixture whose weights depend on M. In the real tasks facts interact strongly, through proofs and context, and Section 10 shows that $\theta_f(M)$ can change sign with M.
2. **Exchangeability** (Approximation 1): $\theta_f = +\beta_C$ for controller facts and $-\beta_A$ for native facts.

An agent holding n_C controller and n_A native facts then has log-odds $x = h_0 + \beta_C n_C - \beta_A n_A$. Here $h_0 = \ln \frac{1/3}{2/3} = \ln \frac{1}{2}$ is the log-odds of the uniform prior over three answers. Probability matching makes the posterior the vote probability, $P(\text{vote } A2) = \sigma(x)$, with $\sigma(x) = 1/(1+e^{-x})$. The mean-field step evaluates this at the average counts $M_\alpha = F_\alpha m_\alpha$:

$$\boxed{\Phi_2 = \sigma(x), \qquad x = h_0 + \beta_C M_C - \beta_A M_A} \tag{12}$$

The mean-field bias. The population's vote probability is $\mathbb{E}[\sigma(x_i)]$, not $\sigma(\mathbb{E}x_i)$. To second order,

$$\mathbb{E}[\sigma(x_i)] \approx \sigma(\bar{x}) + \tfrac{1}{2}\,\sigma''(\bar{x})\,\mathrm{Var}(x_i), \qquad \sigma'' = \sigma'(1-2\sigma). \tag{13}$$

If facts are held independently, $\mathrm{Var}(x_i) = \beta_C^2 F_C m_C(1-m_C) + \beta_A^2 F_A m_A(1-m_A)$. Below the tipping point ($\bar{x} < 0$), $\sigma'' > 0$, so Eq. (12) *underestimates* wrong votes in a heterogeneous population.

Physical reading. Each agent is a non-interacting Ising spin in a local field x_i, with logistic (Glauber) response:

- h_0 is a uniform external field;
- each held fact contributes a field quantum $\pm\beta$;
- the single-spin susceptibility is $\sigma'(x) = \sigma(1-\sigma)$, maximal (1/4) at the tipping point $x = 0$.

Agents never observe each other's votes. The only coupling between spins runs through the shared reservoir of facts, so the theory is mean-field in the literal sense of a global field. One fact multiplies the odds by e^{β}; at $\beta = 0.5$, by 1.65.

Lag. Eq. (12) is the probability of the next vote. The last-vote observable obeys $\dot{\Delta}_v = g - \Delta_v$, with $g = \Phi_2^c - \Phi_2^s$. The model reports g, which equals Δ_v when votes adapt fast.

7 Reposting: Equation 4

Choice model. At each action the agent posts one item, chosen in proportion to weights:

- each held class-α fact has weight $\kappa_\alpha g_\alpha$;
- a background option has weight κ_0. It stands for neutral facts outside both classes and keeps the posting weight positive when the classes are empty.

For an agent with counts (n_A, n_C),

$$\Pr[\text{post class } \alpha] = \frac{w_\alpha}{w_A + w_C + \kappa_0}, \qquad w_\alpha = \kappa_\alpha n_\alpha g_\alpha. \tag{14}$$

Gating. The simulator posts only facts that support the agent's vote. The model compresses this into an eligibility probability with one parameter ε:

$$g_C = \varepsilon + (1-\varepsilon)\,\mathbf{1}[v = A2], \qquad g_A = \varepsilon + (1-\varepsilon)\,\mathbf{1}[v \neq A2]. \tag{15}$$

A held fact is eligible with probability ε regardless of the vote. Otherwise it is eligible only if it supports the vote, and the model assumes controller facts support $A2$ and native facts the other answers.

- $\varepsilon = 1$ is neutral reposting.
- $\varepsilon = 0$ is strict gating.

In stored simulator data a held controller fact is eligible about 80% of the time in task003 and 55% in OP.

Closure. Two mean-field steps give the equation:

- the population share of new class-α posts, $\mathbb{E}[w_\alpha/(w_A + w_C + \kappa_0)]$, is replaced by the ratio of averages (the analogue of an annealed average);
- the vote indicator is replaced by its mean Φ_2, independently of the counts.

The result is

$$\boxed{S_\alpha = \frac{w_\alpha}{w_A + w_C + \kappa_0}, \quad w_\alpha = \kappa_\alpha M_\alpha g_\alpha, \quad g_C = \varepsilon + (1-\varepsilon)\Phi_2, \quad g_A = \varepsilon + (1-\varepsilon)(1-\Phi_2)} \tag{16}$$

The bias of the second step. Agents holding more controller facts are more likely to vote $A2$, so $\mathbb{E}[n_C\mathbf{1}[v = A2]] > \mathbb{E}[n_C]\,\Phi_2$. Under gating ($\varepsilon < 1$), Eq. (16) therefore *underestimates* controller reposting. The ChatGPT agent gives the leading correction: the holder-conditioned log-odds exceeds the population mean by $\beta_f(1-m_f)$, plus covariance terms.

What Eq. 4 leaves out.

- Support measured in context: a controller fact can support a non-$A2$ vote. Section 11 shows this matters.
- The preference for facts not just read.
- Abstention.

The closed system and its parameters. Eqs. (5)–(16) form four ODEs for (m_A, m_C, b_A, b_C) in the controlled arm and two in the silent arm, where $m_C = b_C = 0$. The set $\{m_C = b_C = 0\}$ is invariant once $\lambda = 0$: it is the absorbing manifold of the controller-free state. Initial conditions: each agent holds one native fact on average ($m_A = 1/F_A$), and the board starts empty. Table 1 lists the parameters, all fixed before any run.

Table 1: Parameter set P0. "Simulator" means taken from the simulator's rules; "chosen" means a round number of the right order, not measured or fitted.

parameter	value	source	meaning
N/W	24/48	simulator	board turnover per time unit
W, T	48, 30	simulator	window length, controller stop time
h_0	$\ln \frac{1}{2}$	simulator	uniform prior over three answers
F_A, F_C	16, 4	chosen	spread native evidence; concentrated controller evidence
β_A, β_C	0.5, 0.5	chosen	log-odds per fact
κ_A, κ_C, κ_0	1, 1, 0.2	chosen	posting propensities and background
λ_0	1	chosen	controller posting rate (comparable to one agent)
ε	0–1	varied	vote-independent reposting eligibility

8 What the mean-field system implies

8.1 Frozen board and no communication

Frozen board. With b_α fixed after T, Eq. (5) is linear. Memories relax at rate $r+\gamma$ to the Langmuir values of Eq. (6). Influence is sustained because the frozen board still carries the controller's facts. The stop-time effect is retained exactly only if memories were already stationary at T.

No communication. With $r = 0$ and no posting after T, $m_\alpha(T+s) = m_\alpha(T)e^{-\gamma s}$, so in each arm $x(s) - h_0 = e^{-\gamma s}(x(T) - h_0)$. Both arms return to the prior, and $g \to 0$. Differentiating $g = \sigma(x_c) - \sigma(x_s)$,

$$\frac{dg}{ds}\Big|_{s=0} = -\gamma\Big[\sigma'(x_c)(x_c - h_0) - \sigma'(x_s)(x_s - h_0)\Big]. \tag{17}$$

This can be positive: the silent arm's own evidence, which held it below the prior, fades first. The model shows a transient rise at $\rho = 0.75$ (from 0.027 to 0.063), and so do the simulator's symmetric setups.

8.2 The reading reversal

Let $x_s = h_0 - \beta_A M_A^S$ be the silent arm's log-odds. The controller shifts the controlled arm by the *net displacement*

$$\Delta x = \beta_C M_C - \beta_A\,(M_A^C - M_A^S), \qquad x_c = x_s + \Delta x, \qquad g = \sigma(x_s + \Delta x) - \sigma(x_s). \tag{18}$$

The second term of Δx is the native evidence crowded out of memory by the controller's facts, through the shared board and the shared reading budget q. Differentiating with respect to q, with $\partial_q x_s = -\beta_A \partial_q M_A^S$, gives the exact identity

$$\partial_q g = \underbrace{\sigma'(x_c)\,\partial_q \Delta x}_{\text{exposure}} + \underbrace{\beta_A\,\partial_q M_A^S\,\big[\sigma'(x_s) - \sigma'(x_c)\big]}_{\text{sensitivity}}. \tag{19}$$

Signs, and when they hold. The signs are conditional.

- **Exposure term.** Where reading increases the net displacement, it favours control.
- **Sensitivity term.** If $\beta_A > 0$, reading increases native evidence, and the controlled operating point is more sensitive than the silent one (for example $x_s < x_c \le 0$), then this term protects the swarm. Reading pushes the whole population down the flat tail of the sigmoid.

Short memory keeps M_A small, and exposure wins. Long memory lets M_A grow, and sensitivity wins. For small Δx, $\sigma'(x_c) - \sigma'(x_s) \approx \sigma''(x_s)\Delta x$, so

$$\partial_q g \approx \sigma'(x_s)\Big[\partial_q \Delta x - (1 - 2\Phi_s)\,\beta_A\,\Delta x\,\partial_q M_A^S\Big]. \tag{20}$$

This is a competition the model demonstrates in its working regimes, not a universal sign theorem.

Results (P0). $D(3 \to 6) = +0.003$ at $\rho = 0.05$ and -0.005 at $\rho = 0.5$. The crossover falls with q: it lies at 0.2–0.3 for $3 \to 6$, 0.1–0.2 for $6 \to 9$, and 0.05–0.1 for $9 \to 12$ (Fig. 2). The reversal survives in 16 of 22 single-parameter changes.

It needs none of the following:

- **a proof variable.** Adding the probability z of holding a complete proof, with exact co-retention moments (Section 12), leaves the crossover in place. In the simulator only 1% (task003) and 3% (OP) of agents hold a proof at the crossover.
- **selective reposting.** It is present at every ε.
- **concentrated controller evidence.** It survives $F_C = F_A$, which refuted the saturation mechanism I had predicted in advance.

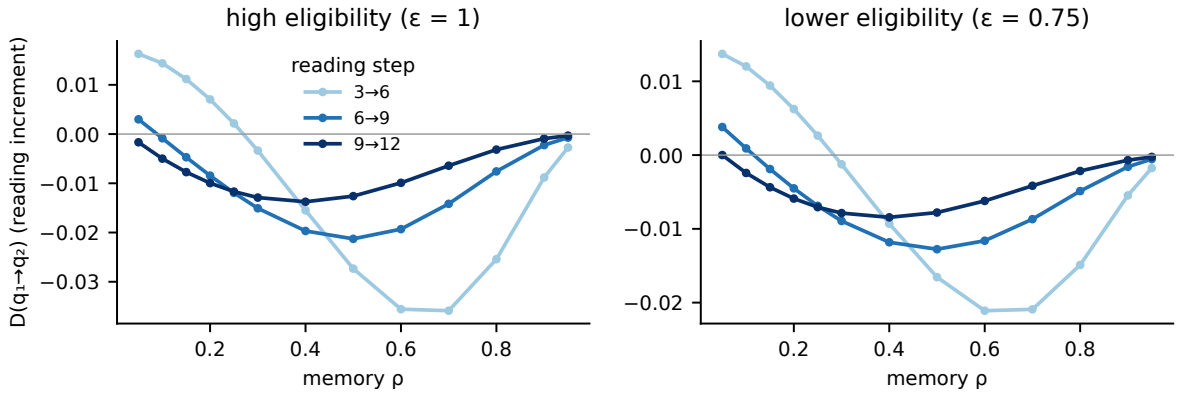


Figure 2: Reading reversal in the mean-field model (P0), for two gatings. Each curve is D for one reading step.

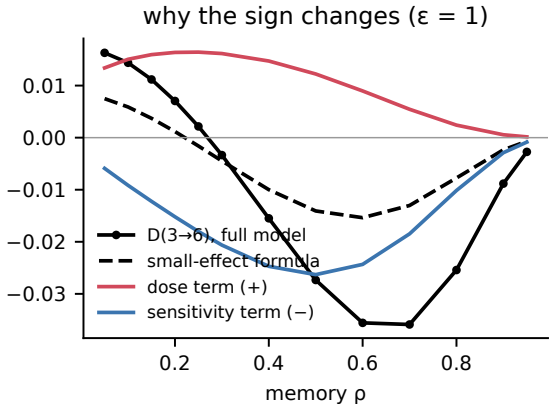


Figure 3: The two terms of the reversal, compared with the full model's reading increment.

Where the crossover sits: counter-evidence. The sensitivity term scales with β_A. In the simulator, we fitted logit $P(A2 \mid M)$ by least squares on class counts, over 72,000 stored memories per setup. This gives an effective β_A:

- OP symmetric: 1.14;
- task003 symmetric: 0.67;

- OP no-proof: -0.07;
- task003 no-proof: -0.17.

This order is the exact reverse of the order of the setups' crossovers, as predicted in advance (Fig. 4). β_A is a context-dependent regression summary, not an intrinsic constant. Inserting the fitted values into the model reproduces the symmetric/no-proof contrast but swaps the two no-proof setups.

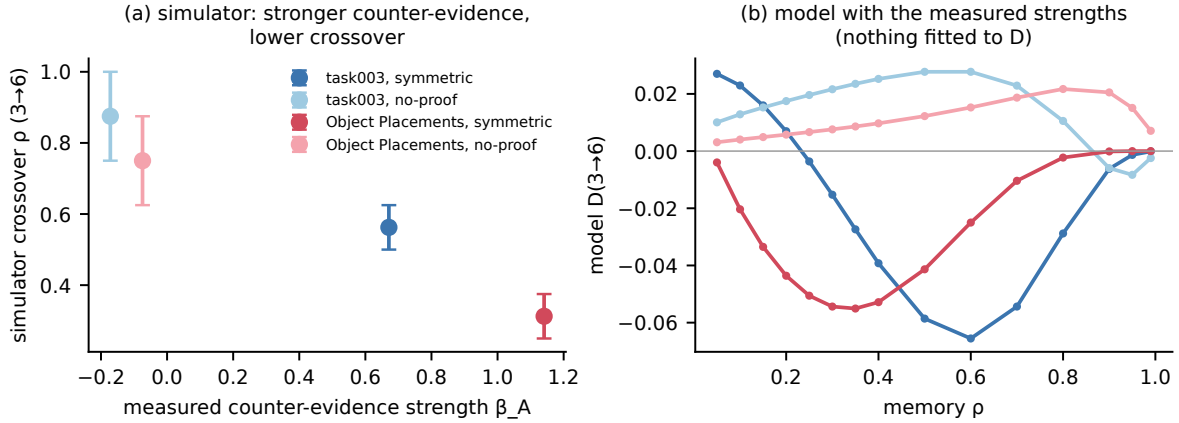


Figure 4: **(a)** Simulator crossovers ordered by the measured counter-evidence strength β_A. **(b)** The model with the measured strengths.

8.3 Persistence: a two-stage invasion threshold

After the stop ($\lambda = 0$), consider a rare controller fact near the controller-free manifold. Write its dynamics in two stages:

- H is the number of its holders;
- B_f is the number of its public copies.

On a frozen background,

$$\dot{H} = a\, B_f - \gamma H, \qquad \dot{B}_f = \pi H - \delta_B B_f. \tag{21}$$

In the fact-flow model the coefficients are:

- $a = Nq/W$: N agents act at rate 1, and each picks a given copy with probability q/W;
- $\pi = \kappa_C g_C / H_{\text{post}}$: a holder's share of posting, with $H_{\text{post}} = \kappa_A M_A g_A + \kappa_0$ the posting weight of all other options;
- $\delta_B = N/W$: the rate at which a copy is pushed out of the window.

The trace of the 2×2 matrix is negative. Its determinant is negative, giving one growing direction, if and only if

$$\boxed{\mathcal{R}_C = \frac{a\,\pi}{\gamma\,\delta_B} = \frac{q\,\kappa_C\,g_C}{\gamma\,H_{\text{post}}} > 1} \tag{22}$$

Reading $\mathcal{R}_C$. It is the basic reproduction number of a two-stage (indirectly transmitted) epidemic. One remembered copy lives $1/\gamma$; during that time its holder posts it at rate π. Each post lives $1/\delta_B$ on the board, during which it is read at rate a. The product is the number of new holders per holder.

Scope. $\mathcal{R}_C$ is a local invasion criterion, computed on a frozen background:

- above threshold, a finite swarm can still lose the fact by extinction (Section 9);
- the threshold does not fix the global nonlinear state;
- sustained influence also requires a circulating state to exist.

Results. Strict gating gives $g_C = \Phi_2$, which is small, so $\mathcal{R}_C \approx 0.2$ and nothing persists. Neutral reposting gives a stable mixed state, an endemic equilibrium, with influence retained. Persistence needs neither bistability nor selection; it does need $\mathcal{R}_C > 1$ (Table 2, Fig. 5).

Table 2: Mean field (P0): retained fraction $g(130)/g(30)$ and $\mathcal{R}_C$ at the stop. Retention is near zero whenever $\mathcal{R}_C < 1$.

ε	0	0.25	0.5	0.75	1
$\rho = 0.25$: retained ($\mathcal{R}_C$)	0.00 (0.20)	0.00 (0.57)	0.00 (0.96)	0.32 (1.37)	0.65 (1.79)
$\rho = 0.75$: retained ($\mathcal{R}_C$)	0.00 (0.03)	0.00 (0.66)	0.29 (1.33)	0.63 (2.03)	0.75 (2.76)

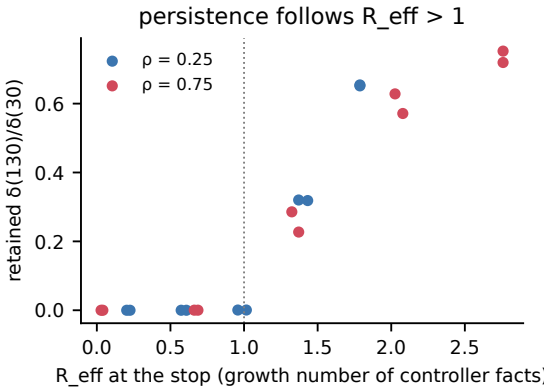


Figure 5: Retained influence against $\mathcal{R}_C$ in the mean-field model.

Where the mean field fails.

- Its controller effect is 3–10 times smaller than the simulator's ($g \approx 0.07$–0.09).
- It reaches its post-stop plateau within 5–9 time units, with no slow fading.

Both failures are finite-size effects.

9 Finite size: copies per fact

The finite model. The same rules, played out with discrete agents. No Bayesian engine is involved:

- $N = 24$ agents, each with a binary memory over $F_A = 16$ native and $F_C = 4$ controller facts. Agent i starts with native fact i mod 16.
- Time steps of $dt = 0.05$:
  - each held fact is forgotten with probability $1 - \rho^{dt}$;
  - each agent acts with probability dt.
- An acting agent:
  - reads q posts with replacement from the 48 newest posts that are not its own;
  - votes $A2$ with probability $\sigma(h_0 + \beta_C n_C - \beta_A n_A)$ on its own counts;
  - posts one item with the weights of Section 7, gated by its own sampled vote.
- The controller posts a uniformly chosen controller fact with probability $\lambda_0\, dt$ per step until T.

A fact is *lost* when it is absent from all memories and all readable windows. Since there is no reintroduction, that state is absorbing. The model measures Δ_v, from 400 paired episodes per point.

The system size is copies per fact. Table 3 gives the stored simulator values. There are only 1.5–12 copies per fact, so each fact's copy number is a small-population birth–death process with an absorbing state at zero. Moreover, at the mean-field state each native fact is close to its own threshold, $\mathcal{R}_f \approx 0.8$–$1.1$: a near-critical branching process.

Exact lineage results (ChatGPT agent).

- **Critical birth–death lineage** (birth and death rate both 1), started from c copies:

$$\Pr[\text{extinct by } t] = \left(\frac{t}{1+t}\right)^c, \qquad \mathbb{E}[n(t)] = c. \tag{23}$$

Table 3: Copies per fact in the simulator (stored runs, $q = 6$, $t = 30$, controlled arm).

cell	native: on board / holders	controller: on board / holders
task003 no-proof $\rho = 0.25$	9.0 / 7.7	7.6 / 7.2
task003 symmetric $\rho = 0.75$	4.4 / 12.2	4.0 / 11.0
OP no-proof $\rho = 0.25$	6.2 / 6.5	5.6 / 6.1
OP no-proof $\rho = 0.75$	4.3 / 11.4	3.6 / 10.7

The mean is conserved while lineages die. With F facts sharing a fixed total of Fc copies, the expected number of surviving facts is $F[1-(t/(1+t))^c]$, so spreading the same total over more facts loses a larger fraction of them.

- **Bounded neutral copying** (a Moran process with K slots) gives the same lesson, exactly:

$$\mathbb{E}[\text{surviving facts}] = F\,[1 - P_t(c, 0)], \tag{24}$$

where $P_t = e^{tQ}$ is the transition matrix of one fact's copy number.

Mean copy numbers, which are all a mean field tracks, cannot determine surviving diversity.

Result 1 (asymmetric extinction)**.** *With P0 parameters unchanged and nothing fitted:*

- ***Diversity collapse.*** *At $t = 30$, 1.6 of 16 native facts still circulate at $\rho = 0.1$, and 11.5 at $\rho = 0.9$. All 4 controller facts survive.*
- ***Effect size.*** *The controller effect rises from 0.07–0.09 (mean field) to 0.31–0.41, a magnitude comparable to selected simulator conditions. The model's reading, posting and controller policies differ from the simulator's, so this is not a matched validation.*
- ***The reversal remains,*** *with the crossover at $\rho \approx 0.5$–0.6.*
- ***Lasting damage.*** *Removing the controller's facts at T leaves a residual effect of $+0.036$.*
- ***Approach to the mean field.*** *Replicating the swarm $R = 1, 2, 4$ times, with the facts unchanged, gives $G = 0.31, 0.21, 0.13$.*

Mechanism. The swarm's evidence is spread over many facts with few copies each, and these die out. The controller recycles a few facts with many copies, and these survive. Extinction removes the swarm's protection but not the controller's push. In a control spreading the same total controller evidence over 16 facts, the amplification over the mean field fell from 4.2× to 2.4×.

The number of surviving facts. Two analytic approximations from mean-field rates bracket it at every memory computed:

- independent two-type branching (holder and board) underestimates it: 0.12–9.4 surviving facts;
- neutral drift at conserved total overestimates it: 3.6–16;
- the finite model lies between: 3.0–12.0.

This is an observed bracket, not a proven bound.

Gating makes persistence fade.

- With neutral reposting, the finite model retains 0.91–0.99 of its effect after 100 time units.
- With vote-gated reposting ($\varepsilon = 0.5$), it retains 0.55 and 0.30 and is still falling (Fig. 7).

Controller lineages lose their reposters and go extinct one at a time. In this model, fading happens through loss of exposure.

10 Heterogeneity of the evidence response

Equation (11) gives the exact persuasion of adding a fact to a memory:

$$\Delta P(M, f) = \sigma\big(x(M) + \theta_f(M)\big) - \sigma\big(x(M)\big). \tag{25}$$

Approximation 2 replaces $\theta_f(M)$ by $+\beta_C$. Measurements on stored memories test this.

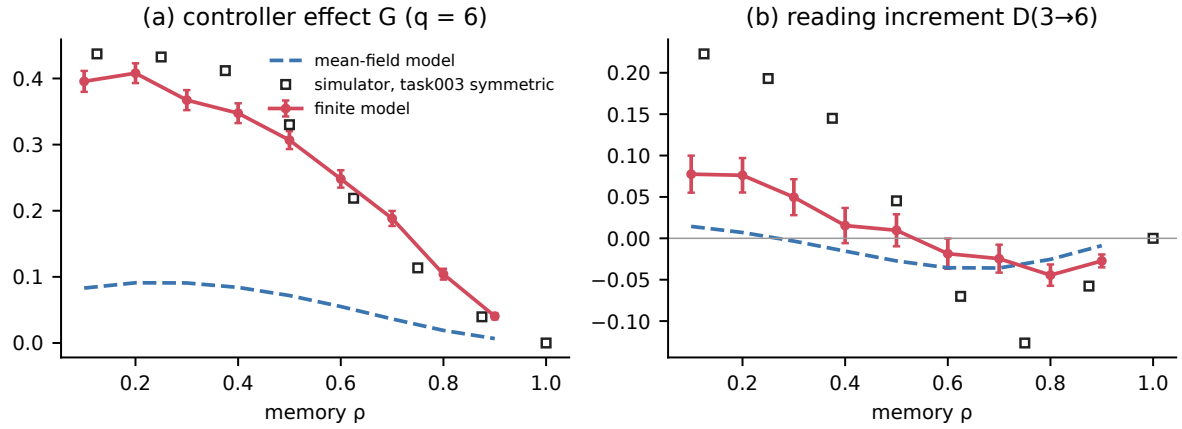


Figure 6: Finite against mean-field model. The simulator (task003 symmetric) is shown for reference only.

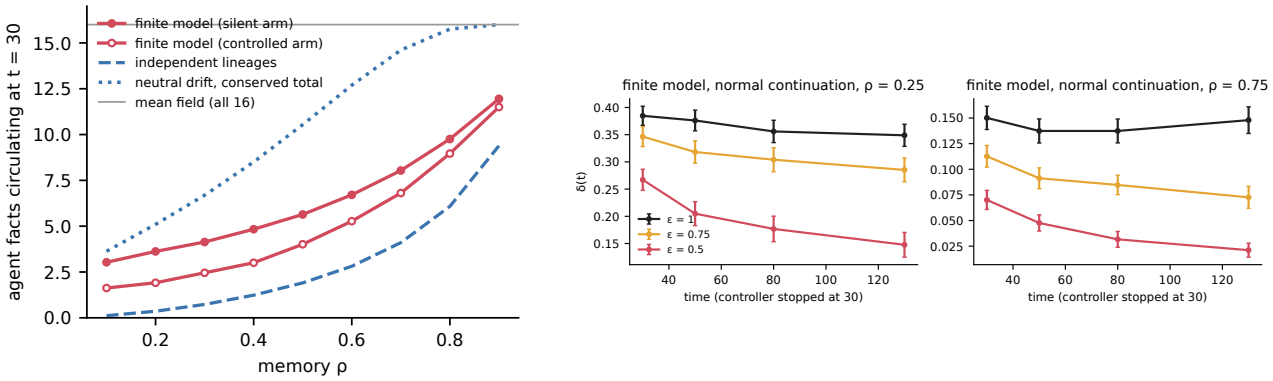


Figure 7: Left: native facts circulating at $t = 30$ in the finite model, with the two analytic approximations. **Right:** vote gating in the finite model turns flat persistence into slow fading.

- **Pre-registered.** The tested count and selected-motif parametrisations do not predict ΔP on held-out episodes ($R^2 \approx 0$). This shows that those parametrisations fail, not that every count or combination model must.
- **Post-hoc.** The identity of the controller fact explains $R^2 = 0.38$–0.78.
- **ChatGPT's extension,** prompted by that result. With flexible per-fact baselines and native-identity controls, pair and triple terms reduce held-out error by a further 9–26% in all four setups.
- **Sign changes.** The same controller fact can push towards or away from $A2$ in memories with equal counts. This is a sign change of $\theta_f(M)$ with context, which no fixed per-fact strength can produce.

Population susceptibility. The linear response of the population to the controller facts it actually encounters is

$$\mathcal{S}(t) = \mathbb{E}_{M\sim\mu_t}\,\mathbb{E}_{f\sim w_t}\big[\Delta P(M, f)\big], \tag{26}$$

with μ_t the distribution of memories and w_t the mix of controller facts on the board. It is an *observable*, not a state variable with its own law. It falls by 36% and 80% in OP no-proof after the stop and stays within ±5% in task003 (Fig. 8).

11 Fading: selection among circulating controller facts

What the simulator shows. In OP no-proof:

- the controller's board share b_C is flat after the stop (-1%, $+5\%$);
- the belief gap Δ_p falls 59% and 89% ($\rho = 0.25$, 0.75), and $\mathcal{S}$ falls 36% and 80%.

So in OP the effect fades *at constant exposure*, unlike the finite model's gated fading. Two further measurements:

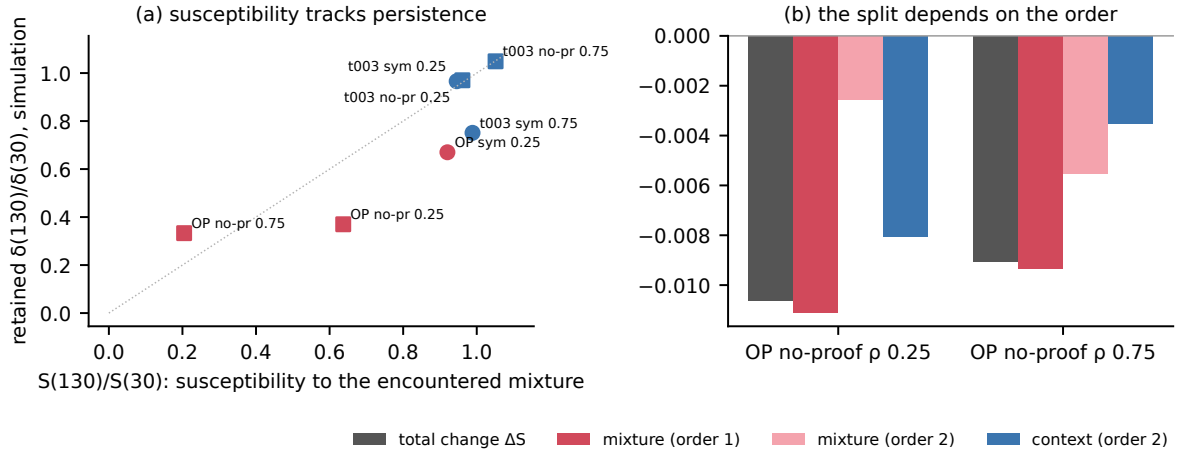


Figure 8: (a) Change in susceptibility against retained effect in the simulator. **(b)** Splitting the change into "mixture" and "context" parts depends on the order of the split; the interaction is large.

- **A frozen-strength probe.** Pairing the stop-time distribution of log-odds strengths with the later operating points does not reproduce the fall. This supports a change in conditional response, though it does not exclude every account based on correlations between operating point and context.
- **The direction of the push.** In task003, 86–99% of encountered (agent, controller fact) pairs push toward $A2$; in OP, only 53–70%.

Coherence. Let c_f be the share of contexts, in a stated reference distribution (silent-arm memories at $t = 30$), in which adding f raises $P(A2)$. It measures sign consistency and ignores magnitudes and zero-effect contexts.

Selection model. The fact-flow model is extended by tracking each controller fact separately. With probability c_f, a fact moves its holder's log-odds by $+\beta_C$, and otherwise by $-\beta_C$, so

$$x = h_0 - \beta_A M_A + \beta_C \sum_f m_f\,(2c_f - 1). \tag{27}$$

Support-based reposting gives the eligibility

$$g_f = \varepsilon + (1-\varepsilon)\Big[c_f\,\sigma(x+\beta_C) + (1-c_f)\big(1-\sigma(x-\beta_C)\big)\Big]. \tag{28}$$

Derivation of Eq. (28).

- With probability c_f the fact pushes its holder toward $A2$. It then supports an $A2$ vote, cast with probability about $\sigma(x+\beta_C)$.
- Otherwise it supports a non-$A2$ vote, cast with probability about $1-\sigma(x-\beta_C)$.

The holder-conditioned shift $\pm\beta_C$ is a crude closure. The exact own-fact shift is $\beta_f(1-m_f)$, plus correlations.

The sign of selection. Using $1-\sigma(y) = \sigma(-y)$,

$$\frac{\partial g_f}{\partial c_f} = (1-\varepsilon)\big[\sigma(x+\beta_C) - \sigma(\beta_C - x)\big], \tag{29}$$

which has the sign of x. When most agents do not vote $A2$ ($x < 0$), selection favours the *least* coherent facts: the majority recirculates controller facts in the contexts where they support its own answer. This is frequency-dependent selection whose sign is set by the sign of the global field. If all coherences and magnitudes are equal, the model has no coherence-based selection.

A Price equation for mean coherence. Let $\bar{c} = \sum_f b_f c_f / b_C$ be the mean coherence on the board, with $b_C = \sum_f b_f$. After the stop, $\dot{b}_f = \frac{N}{W}(S_f - b_f)$. Differentiating $\bar{c}$ gives exactly

$$\dot{\bar{c}} = \frac{N}{W}\,\frac{S_C}{b_C}\,\big(\bar{c}_{\text{new}} - \bar{c}\big), \qquad \bar{c}_{\text{new}} = \frac{\sum_f S_f c_f}{\sum_f S_f}, \tag{30}$$

where S_C is the probability that a new post carries a controller fact. Mean coherence relaxes toward the coherence of what is being reposted. The equation matches the model's numerical derivative to four digits. It is a balance law, not a closed equation:

- assuming $m_f \propto b_f$ gives a drift proportional to $\mathrm{Var}_b(c)$, which overestimates the drift 5–700 times, because each fact's spread saturates;
- the ChatGPT agent built two admissible states with equal $(b_C, \mathcal{S}, \bar{c})$ but different coherence variance, whose $\mathcal{S}$ and $\bar{c}$ drifts differ. So no exact universal closure in these three variables exists on the model's state space. An approximate description on the trajectories actually reached is not excluded.

Pre-registered outcomes (P0; ambivalent set $c = (0.9, 0.7, 0.5, 0.3)$ against coherent set $c = 0.97$):

- **Partly met.** $\bar{c}$ falls whenever reposting depends on the vote. The effect fades beyond exposure by 0.15 at $\varepsilon = 0.5$, but only 0.06–0.08 at $\varepsilon = 0.75$ (threshold 0.1).
- **Met.** The coherent set fades only as much as exposure.
- **Met.** $\varepsilon = 1$ gives no selection.
- **Inconclusive.** The predicted flip under an $A2$ majority: the run did not create such a majority.
- **Failed.** Coherence measured on the whole controller pool does not separate the tasks: task003 no-proof has the most spread-out pool.

Post-hoc evidence.

- Weighting world coherence by the controlled arm's board shares separates the tasks: 0.77–0.89 in task003, unchanged after the stop, against 0.46–0.62 in OP, falling (Fig. 9b). The controller circulates coherent facts when its pool has them.
- Within episodes, surviving controller facts were *more* persuasive than lost ones in task003, and *less* persuasive in OP no-proof at $\rho = 0.75$ (-0.026 ± 0.004).
- ChatGPT's pre-registered closure test: adding $\bar{c}$ to $(b_C, \mathcal{S})$ improves forecasts materially only in OP no-proof $\rho = 0.75$ (+5.5% quadratic ridge; +3.8% nearest neighbours, below the 5% threshold).

The coherence of circulating facts was chosen post-hoc. It is neither an intrinsic world parameter nor a validated cross-task mapping.

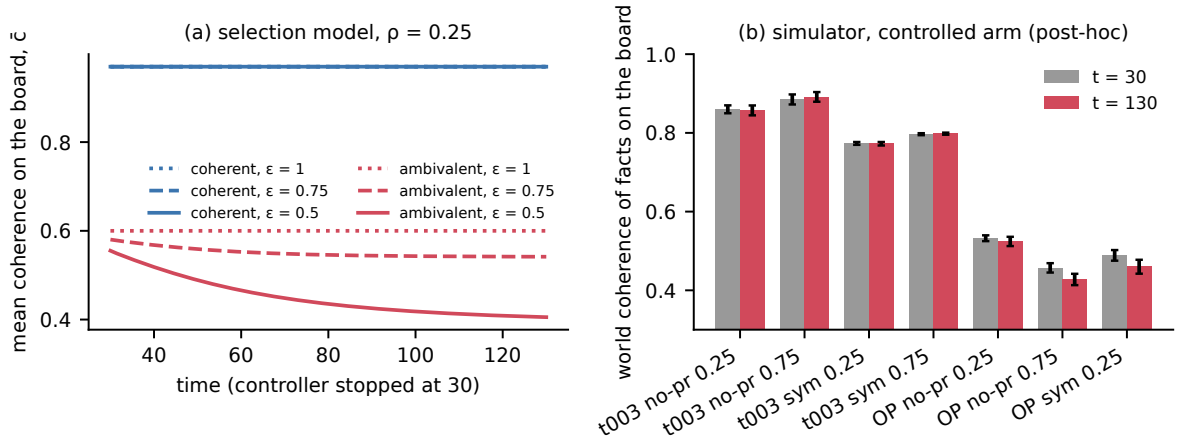


Figure 9: **(a)** Selection model: mean coherence on the board after the stop. **(b)** Simulator (post-hoc): world coherence of the controller facts on the board at the stop and 100 time units later.

Two possible routes to fading.

1. **Selection among circulating controller facts,** as above. It is supported where fading is strongest.
2. **Weakening response at fixed facts** as receiver contexts change. In OP no-proof at $\rho = 0.25$, $\mathcal{S}$ falls 36% with no detectable identity selection and no closure gain from $\bar{c}$. This route has no explanatory law.

These are possibilities within the framework, not a uniquely identified causal partition of individual cells.

12 An alternative closure: exact co-retention (ChatGPT agent)

The ChatGPT agent built an independent model around a different response. A repeated controller claim C competes with information whose meaning requires assembling k interchangeable native facts. Single native facts are inert, and only the full set corrects.

Co-retention. Its central object is the exact co-retention moment h_n: the probability of holding a given set of n native facts. Because a single read can bring several facts at once, held facts are correlated even though each is forgotten independently. Exactly,

$$\dot{h}_n = -n\gamma\, h_n + \sum_{j=1}^{n} P_n(j)\,\big(h_{n-j} - h_n\big), \qquad h_0 = 1, \tag{31}$$

where $P_n(j)$ is the probability that one read brings exactly j of the n facts. Products of single-fact probabilities systematically underestimate h_n.

Reduction. With memory fast compared with the board, the moments become stationary functions of the controller's board share b. Only two variables then evolve:

$$\dot{b} = \nu\{w_C(b) - b[w_A(b) + w_C(b)]\} + J(t)(1-b), \qquad \dot{v} = \Phi(b) - v, \tag{32}$$

with:

- the response $\Phi = p_0 - \beta\tilde{h}_k + \alpha(\tilde{j}_0 - \eta\tilde{j}_k)$, where tildes denote memory just after reading and j_n the probability of holding C together with a given set of n native facts;
- v the mean last vote, $\nu = N/W$, and $J(t)$ the controller's posting rate.

Results.

- The reversal appears at $k = 6$ and fails at $k = 2$ and 3. So a full-proof conjunction cannot be read as the proof size of the task, and it is not our default explanation.
- Persistence appears as invasion of the controller-free state, with growth numbers 1.27 and 1.59 and no bistability. This is a threshold of the same kind as Eq. (22), derived independently.

13 The common framework and its status

The two agents agreed on the following, in a signed statement on the shared blackboard.

1. **Circulation.** Facts are copied between memories and readable windows and are forgotten. The local invasion threshold is $\mathcal{R}_C > 1$.
2. **Evidence response.** The next vote responds to the net log-odds displacement through a sensitivity that depends on the operating point. The reading reversal is the competition of Eq. (19).
3. **Irreversible loss of diversity.** In a finite swarm, the system size is copies per fact. Asymmetric extinction sets the size of the effect and the lasting damage.

The response depends on context, but there is no agreed compact representation of that dependence. Fading has two possible routes (Section 11).

Not explained.

- The size of D: the finite model's is 2–3 times too small.
- The crossover position $\rho^*(q)$.
- The order of the two no-proof setups.
- The scaling with K copies of the project at $K = 4$.
- Route 2 of fading.
- The size of the fading at realistic reposting eligibility.
- The predicted flip of selection under an $A2$ majority.
- Whether rare proof holders matter through their later posts.

Pre-registered and post-hoc.

- **Pre-registered,** with failures reported: the reversal and threshold of the fact-flow model, the finite-model expectations, the β_A ordering, the count/motif test, the susceptibility probe, the fading-chain predictions, and the selection-model predictions.
- **Post-hoc:** the identity result, the survival-by-persuasion comparison, and the coherence of circulating facts.

An out-of-sample test of the coherence parameter, on a setup not used to find it, is left as future work.

Lessons for a physicist. The phenomena are not properties of the inference engine. They emerge from four generic ingredients:

- a two-stage copying process with a reproduction-number threshold;
- a bounded (logistic) response whose susceptibility peaks at a tipping point;
- absorbing extinction at small copy numbers;
- frequency-dependent selection whose sign is set by the global field.

The exact Bayesian structure enters only through effective parameters (β_A, coherence) and through a context dependence of the response that no exchangeable closure captures.

Sources.

- Models: `analysis/theory_claude/phenomenological_fact_flow/` (`fact_flow.py`, `finite_model.py`) and `analysis/theory_claude/fading_chain/selection_model.py`.
- Measurements: `discriminating_test/`, `mixture_probe/`, `fading_chain/`.
- ChatGPT's work: `analysis/theory_chatgpt/`.
- The shared statement: `analysis/theory_blackboard/2026-10-28_claude_18_common_theory_REVISED.md`.